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Contribution of Rating Scales to the Assessment of Executive Functions

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Executive functions play a vital role in the everyday functioning of healthy individuals across the lifespan and have been implicated in a wide variety of clinical conditions. Historically, the assessment of executive functions in clinical and research settings relied on performance-based measures. A number of authors have argued, however, that such measures have limited ecological validity. In response to this limitation of performance-based measures, several rating scales have been developed that seek to gauge a person's or their knowledgeable informant's (e.g., parent or teacher) subjective view of executive functioning in everyday life. In this article we review evidence supporting the use of rating scales of executive function including profiles in clinical populations, biological correlates, relationships to relevant outcome measures such as academic performance, and correlations with performance-based measures. We conclude that performance-based and rating scale measures provide complementary information with respect to a child's executive functions, offering a more comprehensive view than either approach alone.

Key words: ecological validity, executive function, measurement, psychometric, rating scales

INTRODUCTION

Despite the relatively recent introduction of rating scales of executive function in the mid- to late 1990s, a substantial body of literature relies on these behavioral measures as indexes of an individual's everyday real-world self-regulatory functioning. Research across the developmental spectrum shows logical and consistent patterns of findings from these scales in a wide range of clinical conditions as well as associations with diverse behaviors, emotional states, and personality characteristics. Nonetheless, there remain questions as to what exactly these structured behavioral observations actually measure. In part, this stems from their status as the “new kid on the block” versus the “gold

standard” performance measures (i.e., executive function “tests”) that are backed by their own substantial research literature (Anderson, Anderson, Jacobs, & Spencer Smith, 2008), and from an inconsistent pattern of relationships between rating scales and performance measures.

In this article, we discuss the impetus for the development of rating scales of executive function and concerns over their variable relationships to performance measures. We present four lines of evidence that attest to the value of capturing information about an individual's executive function via rating scales of behavior in the everyday real-world context. Although a handful of such rating scales have been developed and a few have been well studied, we focus our arguments on one measure, the Behavior Rating Inventory of Executive Function (BRIEF; Gioia, Isquith, Guy, & Kenworthy, 2000), for several reasons. First, given the brevity of this article and the wealth of literature on point, it is difficult to present data on even one measure, let alone on several measures

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having a substantial literature. Second, the BRIEF is the one measure that covers the full age spectrum from 2 to 90 years. Finally, as authors of the BRIEF, we are intimately familiar with the instrument and the surrounding literature. While we rely on the BRIEF as an exemplar, we believe that the same issues and arguments apply to other rating scales of executive function.

ASSESSING EXECUTIVE FUNCTIONS

The neuropsychological literature on executive function in adults dates back to the late 1800s (e.g., Ferrier, 1886). In contrast, there was little attention paid to the development of executive functions in children until the late 1980s, and only a handful of articles discussed both “children” and “executive function” even by the mid-1990s (Bernstein & Waber, 2007). Much of what we knew then was based on adults with frontal injuries (Stuss & Benson, 1984) and experimental lesion studies (e.g., Diamond, 1988). Even in the earliest articles, however, the use of adult-oriented traditional test-based measurement approaches to understanding executive function in children was questioned (Welsh & Pennington, 1988; Welsh, Pennington, & Grossier, 1991). Welsh et al. noted in their seminal study of children’s performance on standard executive function tasks, “It is misguided to set up adult forms of goal-directed strategizing as the ‘gold standard’ of prefrontal lobe function when one is interested in measuring this neuropsychological domain in children” (Welsh et al., p. 133). Even then, these authors recognized the mistake of considering any given “test” that purported to measure executive function as a “gold standard” in children.

Performance-Based Tests

Many have argued that performance tests of executive function have inherent limitations for assessment regardless of age. The structured and interactive nature of the typical assessment situation reduces demands on executive functions, thereby limiting opportunities to observe critical processes associated with the executive functions (Holmes-Bernstein & Waber, 1990). That is, the examiner provides the structure, planning, organization, and guidance as well as the cueing and monitoring necessary for an individual’s optimal performance; thus, the examiner provides executive control (Kaplan, 1988; Stuss & Benson, 1986). As a result, individuals with substantial executive dysfunction can often perform adequately on tests when the examiner is allowed to structure, cue, and probe for more information, relieving the individual of the need to be appropriately inhibited, flexible, strategic in planning, and goal-directed. Ylvisaker and his colleagues argued that the very requirements of standardization, reliability, and validity for clinic-administered

performance measures restricted the ability to apply assessment findings in the real world (Turkstra, Coelho, & Ylvisaker, 2005). That is, because standardized performance measures must typically be administered in the quiet, nondistracting clinical setting in the same way each time, they fall short of capturing deficits that are present in the everyday, real-world environment.

In the earliest review of the limited empirical literature on executive function in children and adolescents at the time, Pennington and Ozonoff (1996) articulated these and other limitations to using “putative” tests of executive function. They stated that executive functions are molar and tap several executive and nonexecutive components at the same time that can be disrupted in several ways. Differences in cognitive style or ability can affect performance on tests regardless of executive function. The sensitivity and specificity of extant tests is limited such that patients who *should* have executive deficits by virtue of injury or condition often do well on executive function tests while individuals who *should not* have executive deficits often do poorly on such tests. Finally, executive test performance has not been consistently sensitive to frontal versus extrafrontal lesions. Thus, our developing understanding of executive functions in children and adolescents was necessarily skewed by the use of performance measures of executive function.

Rating Scales

Given the central importance of the executive functions to the control of dynamic “real-world” behavior, reliance solely on clinic-based test performance measures potentially can yield a limited, incomplete assessment (Gioia & Isquith, 2004; Silver, 2000). Although performance tests attempt to tap executive functions in explicit and specific ways, they are limited in ecological validity and generalizability to the everyday environment. Burgess (1997) argued that neuropsychological tests alone are inadequate for assessing executive function because they artificially and ambiguously fractionate an integrated system. Performance-based measures tap limited components of the executive function system during a short time frame, and do not tap the integrated, multidimensional, relativistic, priority-based decision making that is often demanded in real-world situations (Goldberg & Podell, 2000).

The challenges to translating performance test findings to the everyday environment led to the development of several rating scale approaches to assessing executive functions. That several separate efforts were under way during the 1990s for both children and adults attests to the consensus that a rating scale approach was a next logical step. Adult-oriented measures, such as the Dysexecutive Questionnaire (Wilson, Alderman, Burgess, Emslie, & Evans, 1996) and Frontal Systems Behavior Scale (Grace & Malloy, 2002), and developmentally

oriented measures such as the BRIEF (Gioia et al., 2000) exploited the efficacy of gathering structured observations of individuals' everyday self-regulatory functioning from them and from observers (e.g., spouses, parents, teachers). Although these behavioral assessments were intended as complementary measures to traditional performance measures, in part to serve as an index of ecological validity for findings in the clinic or laboratory setting, they have become useful measures in their own right as reflected in the substantial body of international literature referencing these measures in a multitude of ways.

Despite their widespread use in clinical and research settings, assessment methods that rely on self-reports and/or informant reports are not without limitations. At the outset, the evaluator has limited control of environmental influences that may affect ratings on behavioral scales. An individual who describes their executive functions as problematic in a highly demanding work environment might describe them as entirely adequate in a less demanding environment. It is also more difficult to parse deficits in specific executive functions via reported behaviors in these everyday contexts than via more narrowly focused performance measures. Executive functions operate in an integrative manner in the everyday context, such as allowing a child to inhibit distraction (e.g., another student whispering to him), thus protecting his working memory to allow holding information (e.g., instructions for a writing assignment and a writing prompt) and a goal (e.g., complete the writing assignment in a timely fashion) in mind, in turn enabling planning (e.g., an approach to the assignment) and organization (e.g., the thesis, three points, and conclusion) of a task while monitoring output (e.g., checking grammar and spelling while writing). Finally, rater perspective must be considered when interpreting rating scales. An individual's emotional state or personality characteristics—or simply whether or not they like or dislike the individual—may influence their ratings. For example, one teacher may dislike a child or, as is sometimes the case, simply dislike completing rating scales, while another teacher may enjoy the same child and appreciate the opportunity to provide input into the evaluation process.

One frequent observation about rating scales and performance measures of executive function is that the two do not correlate often or well. Although recognizing the value of both executive function-rating scales and performance tasks, McCauley and colleagues (2010) noted a problem of mixed correlation findings and offered several possible explanations. One possibility was that the two types of measures assess different aspects of the same underlying construct, such as a behavioral component assessed by rating scales and a cognitive component assessed by performance measures. This was thought, however, to be inconsistent with neuroimaging data showing shared neuroanatomical substrates. The authors

suggested that the performance measures might assess underlying skills, while rating scales assess application of those skills—an explanation that remains to be examined. Alternatively, performance measures might lack ecological validity, though there is a growing body of evidence suggesting that there are meaningful relationships between such tests and real-world functioning such as academics (e.g., Waber, Gerber, Turcios, Wagner, & Forbes, 2006). Finally, McCauley and colleagues raise the possibility that rating scales may not measure executive functions to the extent commonly believed.

EVIDENCE IN SUPPORT OF RATING SCALES

There are several questions we might ask to evaluate whether or not rating scales that purport to capture everyday executive functioning in the real-world environment are actually doing so. First, are rating scales sensitive to executive function deficits in clinical groups? For example, children diagnosed with the Inattentive type of attention-deficit hyperactivity disorder (ADHD-I) might be expected to be described as having difficulties with sustaining working memory, planning, and organizing, while children diagnosed with the Combined type of ADHD (ADHD-C) should also be rated as having problems with impulse control or inhibition. Second, given a presumed neurological basis for executive function deficits, there should be relationships between markers of brain function and ratings on the scales. Third, if rating scales are the more ecologically valid indicators of executive function as often presumed, they should correlate in a meaningful manner with real-world function. And finally, it is worth asking again as an indicator of validity whether or not there are correlations between rating scales and performance measures of executive function.

Profiles of Executive Function

If rating scales of executive function truly measure what they intend to measure, it is reasonable to expect that studies will reveal patterns of ratings or scores that are logical and consistent with expected patterns given a clinical condition. Perhaps the most widely studied clinical group is ADHD. Several studies have found that executive function ratings are sensitive to the presence of ADHD in very young children (Isquith, Gioia, & Espy, 2004; Mahone & Hoffman, 2007), school-aged children and adolescents (Mahone et al., 2002; Mares, McLuckie, Schwartz, & Saini, 2007; Sullivan & Riccio, 2007; Toplak, Bucciarelli, Jain, & Tannock, 2009), and adults (Rotenberg-Shpigelman, Rapaport, Stern, & Hartmen-Maeir, 2008). Both parent and teacher ratings of children's working memory correctly identified approximately 80% of those diagnosed with ADHD (Reddy, Hale, & Brodzinsky,

2011), while ratings of children's inhibitory control helped distinguish between ADHD-I and ADHD-C (Gioia, Isquith, Kenworthy, & Barton, 2002; McCandless & O'Laughlin, 2007). This has led some to question whether or not executive function-rating scales are simply measures of ADHD (e.g., McCauley, et al., 2010). However, these scores show different profiles in other conditions (Gioia et al., 2000) and do not entirely overlap with ADHD rating scales (Gioia et al., 2000). Thus, while executive function-rating scales are sensitive to the symptoms of ADHD, they are complementary to broadband rating scales such as the Behavior Assessment Rating Scale for Children (BASC; Reynolds & Kamphaus, 1992), with the BRIEF providing a detailed picture of executive functioning and the BASC providing information about a range of associated behavioral, social, academic, and mood difficulties (Jarratt, Riccio, & Siekierski, 2005).

Children with autism spectrum disorders (ASDs) often exhibit executive function deficits that, although not causally linked, likely contribute to symptom clusters such as repetitive behaviors and restricted interests. Children with ASDs show difficulty with adjusting to change, which is reflected in elevations on the Shift scale of the BRIEF (Gioia et al., 2002), often accompanied by elevation on the Plan/Organize scale (Kenworthy et al., 2005). Kenworthy and colleagues found that these elevations were significant predictors of severity of autism symptoms (Kenworthy, Black, Harrison, Della Rosa, & Wallace, 2009; Kenworthy, Yerys, Anthony, & Wallace, 2008). Even in high-functioning young adults with sub-clinical autistic traits, self-reported executive functions were remarkable for problems with adjusting to change reflected on the Shift scale (Christ, Kanne, & Reiersen, 2010).

Executive dysfunction is a hallmark of outcome in individuals with traumatic brain injury (TBI). Although a distinctive profile on executive function-rating scales has yet to emerge, difficulties are seen across multiple domains in part related to time since injury and severity of injury. Between publication of the BRIEF and 2010, 16 studies were published showing ratings of more than 1,200 children and adolescents with TBI (Gioia, Kenworthy, & Isquith, 2010) distributed during a wide range of time postinjury, from 1 year (Maillard-Wermelinger et al., 2009), to 5 years (Chapman et al., 2010; Conklin, Salorio, & Slomine, 2008; Donders, DenBraber, & Vos, 2010), and up to 10 years (Mangeot, Armstrong, Colvin, Yeates, & Taylor, 2002; Nadebaum, Anderson, & Catroppa, 2007). A concise review of these studies revealed that 10 studies documented global executive deficits on the overall composite index, 7 showed deficits on the Metacognition Index, and 7 showed deficits on the Behavioral Regulation Index, with Working Memory being the most consistently elevated scale (Gioia et al., 2010).

An important consideration is whether or not behavioral profiles of executive function are distinguishable between clinical groups. In the first study to examine profiles in childhood conditions, parent ratings on the BRIEF were compared between groups of children diagnosed with reading disorders, ADHD-I, ADHD-C, ASD, and moderate TBI or severe TBI evaluated 5 years postinjury (Gioia et al., 2002). A profile analysis showed that children with reading disorders had significantly higher scores compared with matched controls on the Working Memory, Plan/Organize, and Monitor scales, and children with ADHD-I had significantly higher scores than did children with reading disorders and matched controls on the Initiate, Working Memory, Plan/Organize, and Monitor scales. Children diagnosed with ADHD-C were further distinguishable from the ADHD-I group by an elevation on the Inhibit scale, while children with ASDs showed a significantly greater elevation than any other group on the Shift scale. Children with moderate TBI, however, were not distinguishable from controls, while children with severe TBI showed elevations across most scales relative to controls. A more recent rating scale profile comparison between children diagnosed with ASDs with and without ADHD showed different patterns of elevations corresponding to the additional diagnosis of ADHD within ASD (Yerys et al., 2009). This profile analysis method suggests that individuals with clinical conditions that include executive dysfunction, such as ADHD, ASD, and TBI, show expected profiles on a rating scale of everyday executive functioning and that these profiles may be distinguishable between diagnostic groups.

Biological Correlates

A second line of evidence to consider in evaluating whether or not rating scales of executive function actually measure executive functions is the degree to which they are associated with known biological substrates of executive function. Several studies have found associations between neural substrates and everyday executive function as measured by rating scales. In the first such study, Anderson and her colleagues found that BRIEF profiles of children with focal frontal lesions were substantially more elevated than were those of children with diffuse white-matter disruption secondary to early treated phenylketonuria and early treated hydrocephalus, who, in turn, were more elevated than were controls (Anderson, Anderson, Northam, Jacobs, & Mikiewicz, 2002). Children with prefrontal lesions had greater problems with inhibitory control, set shifting, and monitoring, with the highest BRIEF ratings for children with right prefrontal lesions (Anderson, Jacobs, & Harvey, 2005).

Several investigations have examined the relationship between executive function ratings and brain integrity as

measured using magnetic resonance imaging (MRI) scans. Poorer ratings of working memory were associated with smaller frontal-lobe volume in typically developing children (Mahone, Martin, Kates, Hay, & Horska, 2009) and in adults with schizophrenia (Garlinghouse, Roth, Isquith, Flashman, & Saykin, 2010). BRIEF ratings were significantly correlated with cortical thickness (Merkley et al., 2008; Wilde et al., 2012) and white-matter integrity (Wozniak et al., 2007) in children with TBI. In a recent prospective study, shorter corpus callosum length in infancy was associated with poorer executive function ratings at age 4 (Ghassabian et al., 2013). Ratings on the BRIEF also correlated with splenium volume in children with 22q11.2 deletion syndrome (Antshel, Conchelos, Lanzetta, Fremont, & Kates, 2005). In women with breast cancer treated with chemotherapy, difficulties reported on the Initiate scale were associated with reduced frontal-lobe structural integrity in one study (McDonald, Conroy, Smith, West, & Saykin, *in press*), while a study using functional MRI revealed that poorer self-reported executive function was correlated with brain activation during a card-sorting task (Kesler, Kent, & O'Hara, 2011). Together, these studies suggest that there are logical relationships between biological markers of executive function and ratings of everyday executive function.

Ecological Validity

Ecological validity, or the “functional and predictive relation between the patient’s behavior on a set of neuropsychological tests and the patient’s behavior in a variety of real-world settings” (Sbordone, 1996, p. 16), has become an issue of increasing importance as neuropsychology has shifted focus from identifying brain injury to measuring and predicting everyday functioning (Franzen & Wilhelm, 1996). This definition implicates two requirements: First, measures of executive function must bear a functional resemblance to activities in the real world (i.e., verisimilitude), and second, measures must predict or correlate with everyday, real-world functioning (i.e., veridicality). Several authors have argued that rating scales, by their very nature, are high in verisimilitude because they capture observations of those everyday, real-world behaviors (Gioia & Isquith, 2004; Silver, 2000). Recently, studies have begun to explore the veridical relationship between rating scales of executive function and everyday functioning.

In the first study to examine relationships between ratings of executive function and academic performance, Waber and her colleagues (2006) found that teacher ratings on the BRIEF were the best predictor of performance on statewide academic testing relative to executive function performance measures or broadband rating scales. Several studies have found associations between BRIEF scores and reading (Locascio, Mahone, Eason, &

Cutting, 2010) and math (Clark, Pritchard, & Woodward, 2010) skills. Clark and colleagues reported that teacher ratings of 4-year-old preschool children’s everyday executive functions were predictive of math achievement at age 6, over and above performance-based measures of executive function, particularly for working memory, planning, and inhibitory control.

Correlations With Performance Measures

Many studies have not revealed associations between performance measures and rating scales of executive function (e.g., Anderson et al., 2002; Conklin et al., 2008; Mahone et al., 2002; Vriezen & Pigott, 2002), leading some to question whether rating scales that purport to measure executive function are actually measuring something else (McAuley, Chen, Goos, Schachar, & Crosbie, 2010). In a review of 11 studies that compared test findings using a range of performance measures with the BRIEF ratings (McAuley et al., 2010), 3 of the studies found no correlations while 8 found correlations between executive function ratings and one or more measure. A simple count of correlations across the reviewed studies shows approximately 20 null findings between performance measures and rating scales and approximately 14 significant correlations. One might question whether the variability is related to the particular measures or rating scales included in the analysis. For example, 4 of these studies relied on continuous performance tests (CPTs). One study revealed that only response time variability correlated with ratings in one group of children, but in a second group of children, omissions and response time correlated with ratings while response time variability did not (Bodnar, Prahme, Cutting, Denckla, & Mahone, 2007). Mahone and colleagues found that CPT omissions and response variability correlated with ratings of executive function in children with ADHD and/or Tourette syndrome, but commissions and response time did not (Mahone et al., 2002). Another study of children with spina bifida myelomeningocele found no relationships between CPT performance and ratings of executive function (Brown et al., 2008). Thus, the same measure can show different patterns of correlations or none at all with rating scales.

The particular clinical group may account for some of the variability in findings. Ratings of executive function are more likely clinically elevated, for example, in children with epilepsy (Slick, Lautzenhiser, Sherman, & Eyrl, 2006). A recent study of executive function in children with epilepsy found that both the performance measure, the Tower of London (Culbertson & Zilmer, 2000), and the rating scale, the BRIEF, were clinically elevated for similarly substantial proportions of children in the study, but there were no significant correlations between the two measures (MacAllister et al., 2012). Cognitive aspects of

executive function, as measured by the Tower of London, were not related to parent perceptions of executive function. But in a similar study, parent ratings and performance tasks from the Delis-Kaplan Executive Function System (Delis, Kaplan, & Kramer, 2001) were more problematic in children with epilepsy and were correlated (Parrish et al., 2007). This kind of conflicting finding is seen elsewhere as well, such as with comparisons between rating scales and CPTs (McAuley et al., 2010).

DISCUSSION

Performance measures, or tests, to assess executive function were first applied to the study of children's development in the 1980s, with the caveat that there are no "gold standard" tests of executive function in children and that they may be tapping things other than executive functions as we know them in adults (Welsh et al., 1991). These observations, along with increasing calls for attending to the everyday, "real-world" aspects of executive function, served as the impetus for developing executive function-rating scales in the 1990s. Since publication of a handful of these measures, a large body of literature has emerged supporting their use. There are limitations to the validity of rating scales as indicators of executive function, but at the same time, they offer efficiency in data collection and collect useful, ecological information about functioning outside of the lab or clinic setting.

A consistent observation about rating scales of executive function is that they do not consistently correlate with performance measures of the same construct, leading to questions about whether the two types of assessment tools are measuring different aspects or different applications of executive function or even the same construct at all (Anderson et al., 2002; MacAllister et al., 2012; McAuley et al., 2010; Vriezen & Pigott, 2002). Possible explanations include that one or the other type of measure is not actually tapping executive function, that performance measures assess a "cold" cognitive aspect of executive function while rating scales assess a "hot" emotional/behavioral aspect of executive function, that performance measures assess "skills" while rating scales assess application of those skills, that performance measures lack ecological validity, and that rating scales measure a large component of nonexecutive behaviors and are nonspecific.

Despite the inconsistent relationship between performance-based tests and rating scales, several lines of evidence support the validity of rating scales. First, several clinical groups show patterns of everyday executive functioning that are consistent with what might be expected given the clinical condition and that are often distinguishable from others. Thus, there is evidence to suggest that rating scales of executive function capture executive

function characteristics of these groups as they are reflected in everyday functioning. Second, ratings of everyday executive functioning correlate in the expected manner with biological markers. Third, there is a growing body of evidence showing logical concurrent and predictive relationships between real-world functioning such as academic skills and ratings of executive function. Finally, although we are not so fortunate as to find a pattern of strong, logical, and consistent correlations between performance tests of executive function domains and their rating scale namesakes (e.g., working memory, planning, organization, etc.), correlations do exist between the two types of measures. A well-documented set of relationships between like-named performance measures and scales would not only simplify interpretation of test and rating scale findings, but would also obviate the need for one or the other. That is, if tests and rating scales indeed measured exactly the same construct equally well, tests would likely be unnecessary as they are more time-consuming.

We are left, then, with evidence that both putative performance measures of executive function and rating scales capture some aspects of executive function, are related to neurological substrates, and have evidence of association with real-world functioning such as academics, and evidence that they both contribute to an understanding of an individual's functioning. Perhaps a multi-level approach to assessment of executive function is most beneficial by capturing more of the variance than either traditional performance-based measures or rating scales alone. Performance measures attempt to tap specific components of executive function in isolation, while rating scales capture the integrated, multidimensional, relativistic, priority-based decision making that is often demanded in real world situations (Goldberg & Podell, 2000). We continue to advocate for a model of assessment that views the developing child within their environment in a co-constructivist paradigm, or "child-world" model (Holmes-Bernstein & Waber, 1990), that incorporates at least two levels of assessment: specific components defined by performance measures and real-world behavioral manifestations of those specific components measured via rating scales. Both components must then be viewed within the context of the environmental factors that impact the child's function and impact the observations of the child at home and in school.

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